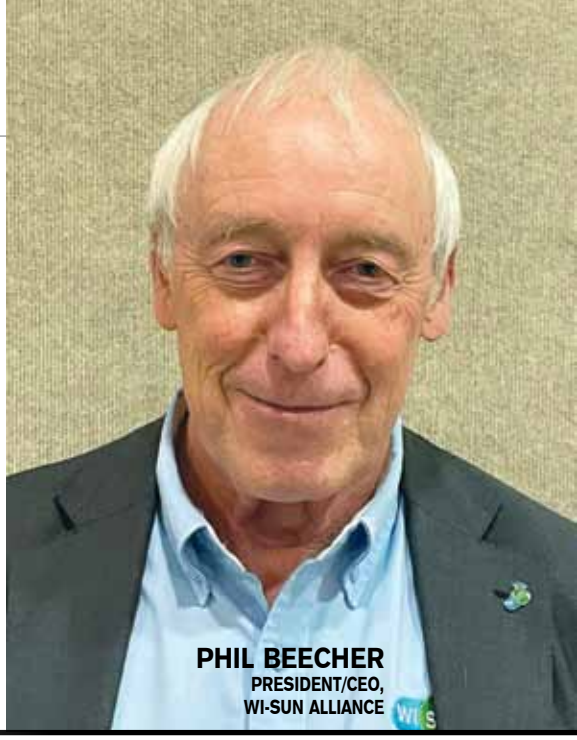
Leaders

"Wi-SUN Is Designed TO PERFORM BETTER As Mesh Networks Become Denser."

Dense networks are where the technology thrives, Wi-SUN Alliance CEO Phil Beecher tells EFY's Nidhi Agarwal, while also discussing post-quantum security, AI, and what open standards mean for Indian developers.



On why Wi-SUN exists

EFY. What technical problems was Wi-SUN designed to solve that existing wireless technologies could not?

Phil Beecher. Wireless Smart Ubiquitous Network (Wi-SUN) was designed to provide interoperable, large-scale wireless mesh networking for critical infrastructure, particularly utility smart metering. When the Wi-SUN Alliance formed, most available solutions were proprietary and offered by individual meter manufacturers, so devices from different vendors couldn't work together. Wi-SUN standardised the communication technology to enable interoperability across vendors while also delivering enterprise-grade security, creating a reliable and secure networking platform for large utility deployments.

On reliability in dense, crowded networks

EFY. How does Wi-SUN maintain reliable communication in large and dense mesh networks with hundreds or thousands of nodes?

PB. Wi-SUN performs better as mesh networks become denser. Frequency hopping lets devices maximise available radio bandwidth while reducing interference. If a point-to-point link or any other point in the network fails, traffic automatically reroutes through neighbouring nodes, ensuring uninterrupted operation.

Wi-SUN also supports multiple border routers (equivalent to data concentrators). In a smart meter network of around 1000 devices, traffic may normally pass through a single border router; if additional border routers are deployed and one fails, communications automatically reroute through another, so data keeps reaching the utility without interruption.

EFY. How does adaptive frequency hopping improve reliability in congested RF environments?

PB. The available spectrum in any radio frequency (RF) environment is limited, and when multiple wireless technologies share the same band, congestion can occur. Wi-SUN technology optimises the use of the available bandwidth not only within the Wi-SUN network but also while sharing the spectrum with other devices. Although heavy traffic limits how many devices can communicate and how much bandwidth is available, Wi-SUN's mesh architecture keeps the network efficient by spreading traffic across neighbouring nodes and channels.

Another important advantage is that mesh nodes can adjust their transmit power. This allows geographically separated groups of devices to communicate with their neighbours without interfering with each other. While adaptive power control is not part of the official Wi-SUN specification, it is commonly implemented by member companies in their products to further improve network reliability.

On power efficiency for field devices

EFY. How does Wi-SUN optimise power consumption for battery-powered field devices?

PB. Wi-SUN uses different modulation techniques depending on the application. Higher data-rate applications use orthogonal frequency-division multiplexing (OFDM), which consumes more power, while battery-powered devices use simpler frequency shift keying (FSK) for very low transmission energy. Duty cycling cuts power further: the radio stays off most of the time and turns on only when communication is needed.

Battery-powered Wi-SUN devices, such as water and gas meters, were